

Creating functions in R

1 Exercise 1

In this Exercise we would like you to write your own functions in R. The general procedure for defining new functions is as follows

```
myfun <- function(argument1, argument2, ...){  
  expression1  
  expression2  
}
```

where `myfun` is the name of the function you are creating, `argument1, argument2, ...` can be any type of R objects you are using as input and `expression` can be any computation and/or already existing function applied to the arguments. By default the result of the last expression is the returned value of the function. Consult the Help manual to learn more about writing your own functions.

Write a function (name it `mysummary`) to compute and display the average and variance of a (vector) sample `x`, using functions such as `mean()`, `var()`, `return()` and `list()`.

2 Exercise 2

The goal of this exercise is to find numerical solutions of a set of estimating (score) equations and to compare them with those obtained with the R built-in functions.

Let's consider the `breast-cancer-wisconsin.txt` dataset. We will use the response `status` (coded 0/1) and only one explanatory variable `nuclei` (plus an intercept).

We consider the logit function as the link between the random part $\mathbb{E}[Y_i|\mathbf{x}_i]$ and the systematic part (linear predictor) $\eta_i = \beta_0 + \beta_1 x_{i1}$.

Based on this, the formula for Newton-Raphson algorithm is:

$$\beta^{(k)} = \beta^{(k-1)} + \left\{ \sum_{i=1}^n \frac{\exp(-\mathbf{x}_i^T \beta^{(k-1)})}{[1 + \exp(-\mathbf{x}_i^T \beta^{(k-1)})]^2} \mathbf{x}_i \mathbf{x}_i^T \right\}^{-1} \times \left\{ \sum_{i=1}^n \left[y_i - \frac{1}{1 + \exp(-\mathbf{x}_i^T \beta^{(k-1)})} \right] \mathbf{x}_i \right\}.$$

- a) Write an R function that computes an update of the estimated parameters, given the previous value, according to the Newton-Raphson algorithm.
- b) Taking the null vector as the starting value for β , perform a few iterations of the Newton-Raphson procedure using a `while()` in the following way:

```
while (max(abs(beta0 - beta1)) > tol & it < maxit) {
  ...
}
```

where `beta0` and `beta1` are the actual and next beta estimations, `tol` is equal to 10^{-8} and `maxit` is 50. Report the estimated β in a table.

- c) Write an R function that computes the log-likelihood of your model.
- d) Use the built-in function `optim()` to solve the maximization problem (with the “BFGS” method). Compare your results with those in **b**).

3 Exercise 3

Consider the profile likelihood on page 111 of the course slides¹:

$$L_1(\theta) = \max_{\delta} L(\theta, \delta),$$

where θ is the parameter of interest of dimension 1, and δ is the vector of the other model parameters. Under this formulation, we construct the following Likelihood Ratio (LR) statistic to test $H_0 : \theta = \theta_0$:

$$\text{LR}(\theta_0) = 2 \left[\log L(\hat{\theta}, \hat{\delta}) - \log L(\theta_0, \hat{\delta}) \right] = 2 \left[\log L_1(\hat{\theta}) - \log L_1(\theta_0) \right] \underset{H_0}{\sim} \chi_1^2,$$

¹Based on a handout by Professor David Patterson, University of Montana <http://www.math.umt.edu/patterson/>

where $\hat{\theta}, \hat{\delta}$ are the maximum likelihood (ML) estimators of θ and δ . The non-rejection region of this test is determined by the condition $\text{LR}(\theta_0) < F^{-1}(1 - \alpha)$, where F is the cdf of the χ^2_1 distribution, or equivalently, all the θ_0 such that:

$$\log L_1(\theta_0) > \log L(\hat{\theta}, \hat{\delta}) - \frac{1}{2} F^{-1}(1 - \alpha) \quad (1)$$

The points within the *Profile Likelihood* CI will be those θ_0 within the non-rejection region of the LR test.

Consider the goose permit data (file `goose.txt`) from Bishop and Heberlein (*Amer. J. Agr. Econ.* 61, 1979), where 237 hunters were each offered one of 11 cash amount (bids) ranging from 1\$ to 200\$ in return of their permits. We use a logit model with $\log(\text{bid})$ as the explanatory variable (in addition to the intercept):

$$\log \left(\frac{\pi_i}{1 - \pi_i} \right) = \beta_0 + \beta_1 \log(\text{bid}_i)$$

where π_i is the probability that a permit holder would accept bid amount bid_i .

- a) Estimate the parameters of the model, extract the log-likelihood with `logLik()` to determine the cutoff value as in (1) for $\alpha = 0.05$ and $\alpha = 0.01$. Compute the profiled likelihood CI for β_1 with the `confint()` function.
- b) (a) Write an R code to compute the profile likelihood function $L_1(\beta_1)$ (which make use of the estimates of the previous point).
(b) Plot the profile likelihood $L_1(\beta_1)$ and the cutoff values.
(c) Compare the results with the CI computed at point a).